Hydrogen in a toroidal cavity: attosecond-field confinement and dynamics

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Abstract

We investigate the ultrafast electronic response of a hydrogen atom confined inside a toroidal geometry whose interior is filled with an attosecond pulse. By coupling the bound electron to a strong, spatially confined, sub-femtosecond field, we track the ensuing wavepacket evolution and resolve the corresponding dynamics in real time. Our results reveal how toroidal confinement and attosecond excitation jointly shape transient population transfer and phase evolution, providing a route to steering and observing atomic motion on its natural timescale within a closed, curved topology.

Introduction

Attosecond science provides direct access to electron motion on its natural timescale and has enabled real-time observation and control of ionization, polarization and coherent wavepacket evolution in atoms and molecules [1]. While most attosecond experiments and models consider fields in free space or in weakly structured environments, there is growing interest in how confinement and curved geometries reshape light–matter interaction by modifying boundary conditions, available states and field distributions [2].

Toroidal structures are particularly appealing in this context. A torus combines closed topology with a non-trivial curvature and supports spatially localized fields that can circulate along the azimuthal direction. When an atom is embedded inside such a geometry, the electron experiences an effective environment that can differ markedly from that of an unconfined system, potentially altering selection rules, transition pathways and phase accumulation.

In this work we study a hydrogen atom placed inside a toroidal region whose interior is filled by an attosecond pulse. Our central goal is to understand how the interplay between (i) geometric confinement imposed by the torus and (ii) the sub-femtosecond driving field governs the ensuing electron dynamics. We focus on the time-resolved evolution of the electronic wavepacket and the associated observables, and we identify characteristic signatures of confinement in the transient response. Beyond providing a minimal model for attosecond dynamics in a closed, curved topology, our results suggest strategies for steering ultrafast atomic motion by shaping both geometry and field.

Results

Theoretical model and mathematical formulation

We model a single active electron in hydrogen interacting with an ultrashort driving field and confined by a toroidal boundary. In atomic units ($\hbar = m_e = e = 1$), the time-dependent Schrödinger equation (TDSE) reads

$$i\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \hat{H}(t)\Psi(\mathbf{r},t), \tag{1}$$

with Hamiltonian

$$\hat{H}(t) = -\frac{1}{2}\nabla^2 - \frac{1}{r} + \hat{V}_{\text{conf}}(\mathbf{r}) + \hat{H}_{\text{int}}(t). \tag{2}$$

Here $-1/r$ is the Coulomb potential of the proton located at the geometric centre of the torus. The confinement is imposed through $\hat{V}_{\text{conf}}(\mathbf{r})$, chosen such that the electron is restricted to the interior of a toroidal region.

Toroidal confinement. We define the torus by its major radius R and minor radius a in cylindrical coordinates (ρ, φ, z), $\rho = \sqrt{x^2 + y^2}$, as the set of points satisfying

$$(\rho - R)^2 + z^2 \leq a^2. \tag{3}$$

To represent an impenetrable boundary (hard-wall confinement), we use

$$V_{\text{conf}}(\rho, z) = \begin{cases} 0, & (\rho - R)^2 + z^2 \leq a^2, \\ +\infty, & (\rho - R)^2 + z^2 > a^2, \end{cases} \tag{4}$$

which enforces the Dirichlet condition $\Psi(\rho, \varphi, z, t) = 0$ on the surface $(\rho - R)^2 + z^2 = a^2$. For numerical work, the hard wall can be approximated by a steep finite barrier, e.g. $V_{\text{conf}} = V_0$ outside the torus with $V_0 \gg 1$.

Light–matter interaction and attosecond pulse. We treat the attosecond pulse in the electric-dipole approximation and length gauge [3],

$$\hat{H}_{\text{int}}(t) = \mathbf{E}(t) \cdot \mathbf{r}, \tag{5}$$

where $\mathbf{E}(t)$ is the (effective) electric field inside the torus. For a linearly polarized pulse along $\hat{\mathbf{e}}$ we write $\mathbf{E}(t) = E(t)\hat{\mathbf{e}}$ with

$$E(t) = E_0\, f(t) \cos\bigl(\omega_0 t + \phi_{\text{CEP}}\bigr), \qquad f(t) = \exp\left[-4 \ln 2\, \frac{(t - t_0)^2}{\tau^2}\right], \tag{6}$$

where E_0 is the peak amplitude, ω_0 the carrier frequency, ϕ_{CEP} the carrier-envelope phase, and τ the full width at half maximum (FWHM) of the intensity envelope.

Attosecond pulse specifications

In the numerical examples below, the pulse duration is chosen in the attosecond regime ($\tau \sim 10^2$–10^3 as), with a central frequency ω_0 in the extreme-ultraviolet (XUV) range to efficiently couple bound and continuum components of the hydrogen wavefunction [1]. The peak amplitude E_0 is selected to remain in the non-relativistic regime while still producing a measurable ultrafast response. The carrier-envelope phase ϕ_{CEP} is retained as a control parameter because it can strongly influence sub-cycle dynamics for few-cycle/attosecond waveforms.

Because the field is assumed to be confined within the toroidal region, we take $\mathbf{E}(t)$ to be spatially uniform inside the torus as a first approximation. More generally, a spatial mode profile $\mathbf{E}(\mathbf{r}, t) = E(t)\, \mathbf{u}(\mathbf{r})$ can be incorporated to capture inhomogeneous cavity/plasmonic fields, provided $\mathbf{u}(\mathbf{r})$ vanishes outside the domain of Eq. (3).

Formation of an attosecond pulse in a torus-filled geometry

In the present model the torus is assumed to be "filled" by an attosecond waveform in the sense that the electromagnetic energy is stored in, and guided by, a toroidal mode whose spatial support is largely restricted to the region defined by Eq. (3). This section summarizes an effective description of how such a field can be formed and parameterized for the TDSE calculations.

Mode picture and field decomposition. We represent the local field inside the torus as the product of a temporal waveform and a normalized spatial mode profile,

$$\mathbf{E}(\mathbf{r},t) = \mathrm{Re}\{\mathcal{E}(t)\,\mathbf{u}(\mathbf{r})\}\,, \qquad \int_{\Omega_{\mathrm{tor}}} |\mathbf{u}(\mathbf{r})|^2\, d^3r = 1, \tag{7}$$

where $\Omega_{\mathrm{tor}} = \{(\rho,\varphi,z) : (\rho - R)^2 + z^2 \le a^2\}$ is the toroidal volume. The complex envelope $\mathcal{E}(t)$ encodes the attosecond pulse, while $\mathbf{u}(\mathbf{r})$ captures confinement and polarization within the cavity.

Spectral synthesis of an isolated attosecond pulse. An isolated attosecond pulse can be viewed as the coherent superposition of a broadband spectrum. Accordingly, we write the complex envelope as

$$\mathcal{E}(t) = \int_{-\infty}^{\infty} A(\omega)\, \exp[i\phi(\omega)] \exp(-i\omega t)\, d\omega, \tag{8}$$

where $A(\omega)$ is the spectral amplitude and $\phi(\omega)$ the spectral phase. Attosecond durations require a large bandwidth and near-linear spectral phase (low chirp). In practice, temporal gating selects a short burst from a longer emission process, while dispersion management flattens $\phi(\omega)$ to approach the transform limit [1, 4, 5].

Effective "filled torus" pulse. To connect this picture to the TDSE, we approximate the intracavity pulse by the analytic form in Eq. (6) and interpret E_0 as an effective peak field inside Ω_{tor}. Finite storage time and losses of the toroidal mode can be incorporated by an additional decay factor, e.g. $E(t) \to E(t)\exp[-(t-t_0)/T_{\mathrm{cav}}]$ for $t > t_0$, where T_{cav} is the cavity lifetime. If a circulating field is required, the mode profile can be chosen azimuthal, for example $\mathbf{u}(\mathbf{r}) \propto \hat{\boldsymbol{\varphi}}\, u(\rho,z)$, so that the local polarization follows the torus.

Atom placement and coupling strength. When the hydrogen atom is placed at a position $\mathbf{r}_a$ within the torus, the relevant driving field entering Eq. (5) is the local value $\mathbf{E}(t) \equiv \mathbf{E}(\mathbf{r}_a, t)$. In the spatially uniform approximation used above, $\mathbf{E}(\mathbf{r}_a, t) = \mathbf{E}(t)$ for all $\mathbf{r}_a \in \Omega_{\mathrm{tor}}$; for inhomogeneous modes the coupling can be tuned by relocating the atom within the cross-section of the torus through the factor $|\mathbf{u}(\mathbf{r}_a)|$.

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Observables. To quantify the attosecond-driven dynamics under confinement we compute, for example, the time-dependent dipole moment

$$\mathbf{d}(t) = \langle \Psi(t)|\mathbf{r}|\Psi(t)\rangle, \tag{11}$$

and the survival probability in the initial state Φ_0,

$$P_0(t) = \left|\langle \Phi_0|\Psi(t)\rangle\right|^2. \tag{12}$$

Energy- and position-resolved densities, $\rho(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2$, are used to visualize wavepacket motion within the toroidal domain defined by Eq. (3).

Solution of the time-dependent Schrödinger equation

We solve the TDSE numerically by expanding the wavefunction in a discrete basis of field-free eigenstates of the confined hydrogen Hamiltonian,

$$\hat{H}_0 \Phi_n(\mathbf{r}) = \epsilon_n \Phi_n(\mathbf{r}), \qquad \hat{H}_0 = -\frac{1}{2}\nabla^2 - \frac{1}{r} + V_{\rm conf}(\rho, z). \tag{13}$$

The time-dependent state is written as

$$\Psi(\mathbf{r}, t) = \sum_{n=1}^{N} c_n(t)\, \Phi_n(\mathbf{r}), \tag{14}$$

where the basis functions satisfy the boundary condition on the torus surface and N is a truncation chosen for convergence.

Coupled-channel equations. Substituting Eq. (14) into the TDSE and projecting onto Φ_m yields a system of coupled ordinary differential equations (ODEs) for the amplitudes $c_n(t)$,

$$i\,\dot{c}_m(t) = \epsilon_m c_m(t) + \sum_{n=1}^{N} V_{mn}(t)\, c_n(t), \tag{15}$$

with dipole coupling matrix elements

$$V_{mn}(t) = \langle \Phi_m|\mathbf{E}(t)\cdot\mathbf{r}|\Phi_n\rangle. \tag{16}$$

The initial condition corresponds to the confined ground state, $c_n(t \to -\infty) = \delta_{n0}$.

Time propagation. Equation (15) is propagated using a unitary scheme. In practice one may use a short-iterative Lanczos/Arnoldi exponential propagator or a split-operator method when the spatial grid representation is preferred [3]. For the coefficient ODEs, a convenient form is

$$\mathbf{c}(t + \Delta t) = \exp\left[-i\,\mathbf{H}_{\rm eff}(t + \tfrac{\Delta t}{2})\,\Delta t\right]\mathbf{c}(t), \tag{17}$$

where $\mathbf{c} = (c_1, \ldots, c_N)^T$ and $[\mathbf{H}_{\rm eff}]_{mn} = \epsilon_m \delta_{mn} + V_{mn}(t)$. Convergence is verified by reducing Δt and increasing N until observables such as $\mathbf{d}(t)$ and $P_0(t)$ change negligibly.

Absorbing boundary for ionization (optional). If ionization or high-energy components reach the numerical boundary, outgoing flux can be removed by adding a weak complex absorbing potential (CAP) near the hard wall, $V_{\text{conf}} \to V_{\text{conf}} - i\eta\, W(\mathbf{r})$ in a thin layer adjacent to $(\rho - R)^2 + z^2 = a^2$, with $\eta > 0$. This prevents unphysical reflections while leaving the bound dynamics essentially unchanged.

Energy spectrum and quantum numbers in toroidal confinement

In free space, hydrogen is characterized by the familiar quantum numbers (n, ℓ, m) and energies $E_n = -1/(2n^2)$ (atomic units), reflecting spherical symmetry and the conserved angular momentum $\hat{\mathbf{L}}^2$ and $\hat{L}_z$. Toroidal confinement breaks full rotational symmetry and removes the degeneracy in ℓ at fixed n. As a result, the eigenstates of $\hat{H}_0$ in Eq. (13) are more naturally labelled by the remaining symmetries of the torus. [3]

Good quantum number: azimuthal number m. Because $V_{\text{conf}}(\rho, z)$ is independent of the azimuthal angle φ, $\hat{H}_0$ commutes with the generator of rotations around the z axis, $\hat{L}_z = -i\,\partial/\partial\varphi$. Therefore, the magnetic quantum number $m \in \mathbb{Z}$ remains a good quantum number, and the eigenstates may be written as

$$\Phi_{\nu m}(\rho, \varphi, z) = \frac{1}{\sqrt{2\pi}}\, \chi_{\nu m}(\rho, z)\, e^{im\varphi}, \tag{18}$$

where ν indexes the remaining degrees of freedom in the (ρ, z) cross-section.

Parity in z. The torus region in Eq. (3) is symmetric under reflection $z \to -z$, and so is the Coulomb potential. Hence $\hat{H}_0$ also commutes with the z-parity operator, allowing classification by $\pi_z = \pm 1$ (even/odd) according to $\chi_{\nu m}(\rho, -z) = \pi_z \chi_{\nu m}(\rho, z)$.

Effective "radial" and "poloidal" indices. Unlike spherical confinement, there is no unique analytic principal quantum number. We therefore use ν as a compound index that can be interpreted as a pair of node-counting integers in the cross-section, e.g. (n_ρ, n_z), or equivalently as a poloidal-mode index n_p and a radial-mode index n_r, depending on the numerical representation. In all cases the indexing is defined by ordering the discrete eigenenergies $\epsilon_{\nu m}$.

Bound-state energies. The confined bound-state energies are obtained from the eigenvalue problem in Eq. (13):

$$\epsilon_{\nu m} = \langle \Phi_{\nu m} | \hat{H}_0 | \Phi_{\nu m} \rangle, \qquad \epsilon_{\nu m} < 0 \text{ (bound)}. \tag{19}$$

For fixed m, confinement raises (makes less negative) the energies relative to free hydrogen, and splits states that would otherwise be degenerate in ℓ. In the limit $R \gg a$ (thin torus), the spectrum approaches that of an atom in a quasi-cylindrical annular confinement, whereas for large a the free-space hydrogen levels are progressively recovered.

Selection rules under linear polarization. For a linearly polarized field in Eq. (5), the dominant dipole selection rules retain the azimuthal constraint $\Delta m = 0$ for polarization along z and $\Delta m = \pm 1$ for polarization in the (x, y) plane. The remaining "ν" transitions are governed by overlap between the corresponding confined eigenmodes and are typically strongest between states with nearby node structure.

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Observation of ultrafast dynamics from the TDSE and spectrum

The TDSE solution provides the full time-dependent state $\Psi(\mathbf{r}, t)$, from which both real-space dynamics and energy-domain information can be extracted. In the toroidal geometry, the combination of a discrete confined spectrum $\{\epsilon_{\nu m}\}$ and attosecond driving produces wavepacket motion that is naturally resolved in terms of the quantum numbers (m, π_z, ν).

State-resolved populations and channels. Projecting the propagated state onto the field-free eigenbasis yields instantaneous populations

$$P_{\nu m}(t) = \left| \langle \Phi_{\nu m} | \Psi(t) \rangle \right|^2, \tag{22}$$

which identify which confined channels are populated by the pulse. Because m is conserved or changed only according to the dipole selection rules, $P_{\nu m}(t)$ provides a direct route to tracking angular-momentum transfer around the torus axis (via m) and symmetry changes in the cross-section (via π_z). Transient beatings in $P_{\nu m}(t)$ reflect energy splittings $\Delta\epsilon$ in the confined spectrum through characteristic frequencies $\Omega = \Delta\epsilon$ (atomic units).

Real-space visualization of wavepacket motion. The probability density $\rho(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2$ reveals where the electron is localized within the torus. For a toroidal domain it is often convenient to examine cross sections $\rho(\rho, z; t)$ at fixed φ and azimuthal averages

$$\bar{\rho}(\rho, z; t) = \frac{1}{2\pi} \int_0^{2\pi} \rho(\rho, \varphi, z; t)\, d\varphi, \tag{23}$$

which isolate the dynamics in the poloidal cross-section and can be compared across different m-channels.

Dipole traces and spectral fingerprints. The dipole response $\mathbf{d}(t)$ encodes the driven polarization and can be Fourier transformed to obtain a spectral signal

$$\mathbf{D}(\omega) = \int_{-\infty}^{\infty} \mathbf{d}(t)\, e^{i\omega t}\, dt, \tag{24}$$

whose peaks are associated with transition energies between confined levels, $\omega \approx \epsilon_{\nu' m'} - \epsilon_{\nu m}$, subject to the selection rules in m and parity. In this way, the confined energy spectrum and quantum-number structure provide an interpretable map from time-domain oscillations to frequency-domain features.

Time–frequency analysis (attosecond resolution). Because the driving field is sub-femtosecond, non-stationary dynamics can be resolved by time–frequency methods such as a windowed Fourier transform of $d(t)$,

$$D(\omega, t_c) = \int d(t)\, w(t - t_c)\, e^{i\omega t}\, dt, \tag{25}$$

where w is a short gate function. This representation separates prompt, field-driven response from post-pulse free evolution and highlights when particular transitions (identified by $\epsilon_{\nu m}$ differences) contribute most strongly.

Connection to experimentally accessible observables. In a cavity or nanostructured realization, the calculated $\mathbf{d}(t)$ and $\mathbf{D}(\omega)$ relate to emitted radiation and transient absorption signals [2], while the channel populations $P_{\nu m}(t)$ provide a theoretical analogue of state-resolved spectroscopy. Thus, combining the TDSE solution with the confined spectrum and quantum numbers yields a consistent framework for "observing" ultrafast electron dynamics in the torus, both in real space and in spectral measurements.

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Simulation protocol and numerical parameters

This section summarizes a practical simulation workflow for the torus-confined hydrogen atom driven by an attosecond pulse. The steps below are compatible with either (i) a basis-expansion approach as in Eqs. (13)–(17) or (ii) a direct grid-based propagation in (ρ, φ, z).

Geometry and confinement parameters. We specify the torus by (R, a) and implement confinement via the hard-wall condition Eq. (4) or, numerically, a finite barrier V_0 outside the region of Eq. (3). Unless otherwise stated, the hydrogen nucleus is placed at the geometric centre of the torus and the atom is initialized in the confined ground state Φ_0.

Field parameters. The attosecond pulse is defined by $\{E_0, \omega_0, \tau, t_0, \phi_{\mathrm{CEP}}, \hat{\mathbf{e}}\}$ in Eq. (6). For a "filled torus" mode we either assume a uniform field inside Ω_{tor} or use a prescribed spatial profile $\mathbf{u}(\mathbf{r})$ in Eq. (9). Polarization is chosen to control Δm according to the selection rules described above.

Basis construction (eigenstate method). We compute N field-free eigenpairs $(\epsilon_{\nu m}, \Phi_{\nu m})$ of $\hat{H}_0$ subject to toroidal boundary conditions. In practice, the basis is assembled separately for each m (and z-parity) block to exploit symmetry and reduce computational cost. The TDSE is then propagated by solving Eq. (15) with a unitary exponential propagator Eq. (17).

Grid construction (direct propagation option). Alternatively, one may discretize the torus region on a 3D grid and propagate the wavefunction by split-operator or Crank–Nicolson schemes. In that case, the hard wall is enforced by setting $\Psi = 0$ at grid points outside Ω_{tor}, and absorbing layers (CAP or mask functions) are placed adjacent to the boundary when ionization is relevant.

Time stepping and convergence. The time step Δt is chosen to resolve both the carrier oscillation $(2\pi/\omega_0)$ and the fastest bound-state beating frequencies $\Delta\epsilon$ in the confined spectrum. Convergence is assessed by: (i) reducing Δt, (ii) increasing basis size N (or grid resolution), and (iii) varying CAP strength/width, until observables ($P_{\nu m}(t)$, $\mathbf{d}(t)$ and spectra) change negligibly.

Table 1: Example simulation parameter list (edit with your chosen values).

Category	Parameter	Meaning
Geometry	R, a	Torus major/minor radii
Confinement	V_0	Finite-barrier height (if used)
Pulse	E_0	Peak field amplitude
Pulse	ω_0	Carrier frequency
Pulse	τ	FWHM duration
Pulse	ϕ_{CEP}	Carrier-envelope phase
Numerics	Δt	Time step
Numerics	N	Number of eigenstates (basis method)
Numerics	η, layer width	CAP strength and width (optional)

Figure 1: Case I: Time-resolved density snapshots $\rho(\rho, z; t)$ (or azimuthal averages $\bar{\rho}(\rho, z; t)$) illustrating wavepacket motion inside the torus during attosecond excitation.

Recommended outputs and plots. We compute and report: (i) density snapshots $\rho(\rho, \varphi, z; t)$ and azimuthal averages $\bar{\rho}(\rho, z; t)$, (ii) state-resolved populations $P_{\nu m}(t)$ and parity-resolved totals, (iii) dipole traces $d(t)$ and spectra $|D(\omega)|^2$, and (iv) time–frequency maps $|D(\omega, t_c)|^2$ from Eq. (29). These quantities jointly connect real-space motion, quantum-number channel dynamics, and spectral fingerprints of the confined energy level structure.

Simulation results

In this section we summarize representative simulation outputs for the torus-confined hydrogen atom driven by an attosecond pulse. Figures referenced below are placeholders that can be replaced by your simulation images.

Case I: Reference geometry and baseline pulse

We first consider a reference torus geometry (R, a) and a baseline attosecond pulse $(E_0, \omega_0, \tau, \phi_{\mathrm{CEP}})$. Figure 6 shows snapshots of the probability density $\rho(\mathbf{r}, t)$ in the torus cross-section during and after the pulse, revealing the formation and subsequent evolution of an electronic wavepacket confined within Ω_{tor}. The corresponding dipole trace $d(t)$ and spectrum $|D(\omega)|^2$ (Fig. 7) display pronounced oscillations and discrete spectral peaks, consistent with transitions between confined levels $\epsilon_{\nu m}$.

Case II: Geometric variation (R and a)

To isolate the role of confinement, we vary the torus radii. Figure 8 compares dipole spectra for several (R, a) values. As confinement strengthens (smaller cross-section a and/or reduced free volume), level spacings increase and spectral peaks shift to higher frequencies. This geometrically induced spectral tuning is accompanied by changes in wavepacket localization and beating frequencies in the state populations $P_{\nu m}(t)$.

Case III: Pulse control (E_0, τ, and ϕ_{CEP})

Next, we examine how the attosecond waveform controls the confined dynamics. Figure 9 shows representative channel populations $P_{\nu m}(t)$ for different pulse strengths and phases. Increasing E_0 enhances population transfer and can populate higher ν modes, whereas the carrier-envelope phase ϕ_{CEP} modifies sub-cycle timing and interference between excitation pathways, leading to measurable differences in post-pulse coherences and beating patterns. Shorter pulses (smaller τ) broaden the spectrum and can access a wider set of confined transitions simultaneously.

Figure 2: Case I: (Top) Dipole response $d(t)$ and (bottom) spectral intensity $|D(\omega)|^2$. Peaks indicate allowed transitions between confined eigenstates, subject to Δm and parity selection rules.

Figure 3: Case II: Comparison of $|D(\omega)|^2$ for different torus geometries (R, a), illustrating confinement-induced shifts in transition frequencies and changes in spectral weight.

Case IV: Quantum-number-resolved dynamics (m and π_z)

Finally, we resolve the dynamics by the conserved and symmetry-derived quantum numbers. For polarization along z, the response is dominated by $\Delta m = 0$ channels, while in-plane polarization leads to $\Delta m = \pm 1$ pathways. Figure 10 presents an m-resolved decomposition of the dipole (or population) dynamics, highlighting how toroidal symmetry constrains excitation and how parity (π_z) selection influences which modes contribute to the observed spectra.

Parameter dependence and discussion of observables

To connect the theoretical formulation with the simulated signals, we summarize here how the key model parameters control the observed ultrafast dynamics, and which computed observables provide the most direct diagnostics.

Geometry parameters (R and a). The major radius R controls the overall torus topology and available volume, while the minor radius a sets the strength of confinement in the poloidal cross-section. Decreasing a strengthens confinement, typically increasing level spacings $\Delta\epsilon$ and shifting dipole spectral peaks to higher frequencies. Changes in R and a also modify the spatial mode overlap with the cavity field profile $\mathbf{u}(\mathbf{r})$ and therefore change the effective coupling. These effects are observed through (i) the confined eigenenergies $\epsilon_{\nu m}$, (ii) beating frequencies in populations $P_{\nu m}(t)$, and (iii) shifts in $|D(\omega)|^2$.

Pulse parameters (E_0, ω_0, τ, ϕ_{CEP}). The peak amplitude E_0 sets the interaction strength and controls the degree of population transfer and ionization probability. The carrier frequency ω_0 determines which confined transitions are near-resonantly driven, while the duration τ controls bandwidth and thus the number of accessible channels. For few-cycle/attosecond excitation, the carrier-envelope phase ϕ_{CEP} influences sub-cycle timing and interference between pathways, which can be tracked via phase-sensitive post-pulse coherences and time–frequency maps $|D(\omega, t_c)|^2$.

Quantum numbers and symmetry control (m and π_z). The azimuthal quantum number m and z-parity π_z provide a compact description of the response under toroidal symmetry. Changing the polarization direction selects Δm and thus the set of accessible channels. A quantum-number-resolved analysis (e.g. $P_{\nu m}(t)$ summed over ν for each m) clarifies which symmetry sectors dominate the measured dipole response and its spectrum.

How the dynamics is "observed" in practice. Within the TDSE framework, the dynamics is directly visible in (i) real-space densities $\rho(\mathbf{r}, t)$ and cross-sections $\bar{\rho}(\rho, z; t)$, and (ii) projections onto the confined spectrum via $P_{\nu m}(t)$. For comparison with experiments, the most relevant quantities are (iii) the dipole response $d(t)$ and (iv) spectral observables $|D(\omega)|^2$ and $|D(\omega, t_c)|^2$, which correspond to emitted radiation and transient absorption-like signals. Presenting these four views together provides a self-consistent interpretation linking theory (Hamiltonian and symmetries), simulation outputs, and experimentally motivated observables. revealing the formation and subsequent evolution of an electronic wavepacket confined within Ω_{tor}. The corresponding dipole trace $d(t)$ and spectrum $|D(\omega)|^2$ (Fig. 7) display pronounced oscillations and discrete spectral peaks, consistent with transitions between confined levels $\epsilon_{\nu m}$.

Figure 4: Case III: State-resolved populations $P_{\nu m}(t)$ demonstrating control of confined dynamics through pulse amplitude E_0, duration τ, and carrier-envelope phase ϕ_{CEP}.

Figure 5: Case IV: Quantum-number-resolved dynamics. Example of m- (and/or π_z-) resolved populations or dipole contributions under different polarizations, demonstrating symmetry-driven channel selectivity in the toroidal geometry.

Case II: Geometric variation (R and a)

To isolate the role of confinement, we vary the torus radii. Figure 8 compares dipole spectra for several (R, a) values. As confinement strengthens (smaller cross-section a and/or reduced free volume), level spacings increase and spectral peaks shift to higher frequencies. This geometrically induced spectral tuning is accompanied by changes in wavepacket localization and beating frequencies in the state populations $P_{\nu m}(t)$.

Case III: Pulse control (E_0, τ, and ϕ_{CEP})

Next, we examine how the attosecond waveform controls the confined dynamics. Figure 9 shows representative channel populations $P_{\nu m}(t)$ for different pulse strengths and phases. Increasing E_0 enhances population transfer and can populate higher ν modes, whereas the carrier-envelope phase ϕ_{CEP} modifies sub-cycle timing and interference between excitation pathways, leading to measurable differences in post-pulse coherences and beating patterns. Shorter pulses (smaller τ) broaden the spectrum and can access a wider set of confined transitions simultaneously.

Case IV: Quantum-number-resolved dynamics (m and π_z)

Finally, we resolve the dynamics by the conserved and symmetry-derived quantum numbers. For polarization along z, the response is dominated by $\Delta m = 0$ channels, while in-plane polarization leads to $\Delta m = \pm 1$ pathways. Figure 10 presents an m-resolved decomposition of the dipole (or population) dynamics, highlighting how toroidal symmetry constrains excitation and how parity (π_z) selection influences which modes contribute to the observed spectra.

Simulation protocol and numerical parameters

This section summarizes a practical simulation workflow for the torus-confined hydrogen atom driven by an attosecond pulse. The steps below are compatible with either (i) a basis-expansion approach as in Eqs. (13)–(17) or (ii) a direct grid-based propagation in (ρ, φ, z).

Geometry and confinement parameters. We specify the torus by (R, a) and implement confinement via the hard-wall condition Eq. (4) or, numerically, a finite barrier V_0 outside the region of Eq. (3). Unless otherwise stated, the hydrogen nucleus is placed at the geometric centre of the torus and the atom is initialized in the confined ground state Φ_0.

Field parameters. The attosecond pulse is defined by $\{E_0, \omega_0, \tau, t_0, \phi_{\mathrm{CEP}}, \hat{\mathbf{e}}\}$ in Eq. (6). For a "filled torus" mode we either assume a uniform field inside Ω_{tor} or use a prescribed spatial profile $\mathbf{u}(\mathbf{r})$ in Eq. (9). Polarization is chosen to control Δm according to the selection rules described above.

Basis construction (eigenstate method). We compute N field-free eigenpairs $(\epsilon_{\nu m}, \Phi_{\nu m})$ of $\hat{H}_0$ subject to toroidal boundary conditions. In practice, the basis is assembled separately for each m (and z-parity) block to exploit symmetry and reduce computational cost. The TDSE is then propagated by solving Eq. (15) with a unitary exponential propagator Eq. (17).

Figure 6: Case I: Time-resolved density snapshots $\rho(\rho, z; t)$ (or azimuthal averages $\bar{\rho}(\rho, z; t)$) illustrating wavepacket motion inside the torus during attosecond excitation.

Figure 7: Case I: (Top) Dipole response $d(t)$ and (bottom) spectral intensity $|D(\omega)|^2$. Peaks indicate allowed transitions between confined eigenstates, subject to Δm and parity selection rules.

Grid construction (direct propagation option). Alternatively, one may discretize the torus region on a 3D grid and propagate the wavefunction by split-operator or Crank–Nicolson schemes. In that case, the hard wall is enforced by setting $\Psi = 0$ at grid points outside Ω_{tor}, and absorbing layers (CAP or mask functions) are placed adjacent to the boundary when ionization is relevant.

Time stepping and convergence. The time step Δt is chosen to resolve both the carrier oscillation $(2\pi/\omega_0)$ and the fastest bound-state beating frequencies $\Delta\epsilon$ in the confined spectrum. Convergence is assessed by: (i) reducing Δt, (ii) increasing basis size N (or grid resolution), and (iii) varying CAP strength/width, until observables ($P_{\nu m}(t)$, $\mathbf{d}(t)$ and spectra) change negligibly.

Recommended outputs and plots. We compute and report: (i) density snapshots $\rho(\rho, \varphi, z; t)$ and azimuthal averages $\bar{\rho}(\rho, z; t)$, (ii) state-resolved populations $P_{\nu m}(t)$ and parity-resolved totals, (iii) dipole traces $d(t)$ and spectra $|D(\omega)|^2$, and (iv) time–frequency maps $|D(\omega, t_c)|^2$ from Eq. (29). These quantities jointly connect real-space motion, quantum-number channel dynamics, and spectral fingerprints of the confined energy level structure.

Discussion

We have presented a theoretical framework for attosecond-driven electron dynamics of hydrogen confined within a toroidal geometry. The model combines a hard-wall (or steep-barrier) toroidal confinement with a short, intense attosecond waveform that is assumed to be supported inside the torus volume, yielding a tractable setting in which topology, symmetry and ultrafast driving can be studied on equal footing.

Our analysis highlights three central points. First, toroidal confinement breaks the free-space hydrogen degeneracies and replaces the usual (n, ℓ, m) classification by a symmetry-adapted set of quantum numbers in which the azimuthal number m and z-parity remain good labels, while the remaining structure is captured by node-counting indices in the cross-section. Second, the TDSE solution provides a direct route to observing dynamics through complementary views: real-space density evolution, channel-resolved populations $P_{\nu m}(t)$, and frequency- or time–frequency signatures derived from the dipole response. Third, both geometry (R, a) and pulse parameters $(E_0, \omega_0, \tau, \phi_{\text{CEP}})$ offer clear control knobs, allowing confinement-induced spectral tuning and phase-sensitive manipulation of sub-cycle wavepacket motion.

Several limitations should be noted. The present treatment assumes the dipole approximation and a prescribed intracavity field; a fully self-consistent description would couple the electron to a Maxwell solver to determine $\mathbf{E}(\mathbf{r}, t)$ and account for back-action. In addition, the hard-wall boundary is an idealization of a physical torus and can be refined by adopting finite potentials or material-specific boundary conditions. Extensions to multi-electron atoms, effective core potentials, or inclusion of decoherence and temperature effects would further broaden the applicability of the approach.

Despite these simplifications, the framework establishes a transparent link between toroidal symmetry, the confined energy spectrum and attosecond observables, and it provides practical guidance for simulations and for interpreting future cavity- or nanostructure-based measurements of ultrafast electron dynamics.

Figure 8: Case II: Comparison of $|D(\omega)|^2$ for different torus geometries (R, a), illustrating confinement-induced shifts in transition frequencies and changes in spectral weight.

Figure 9: Case III: State-resolved populations $P_{\nu m}(t)$ demonstrating control of confined dynamics through pulse amplitude E_0, duration τ, and carrier-envelope phase ϕ_{CEP}.

Methods

Atomic units and numerical conventions

All calculations are formulated in atomic units ($\hbar = m_e = e = 1$). Time is reported in atomic units unless otherwise stated; attosecond durations are converted using standard factors.

Geometry and confinement

The torus geometry is defined by (R, a) via Eq. (3). Hard-wall confinement is implemented through the Dirichlet boundary condition on $(\rho - R)^2 + z^2 = a^2$, or approximated by a steep finite barrier V_0 outside the toroidal domain.

Pulse definition

The driving field follows Eqs. (5) and (6), optionally combined with a spatial mode profile as in Eq. (9). The numerical pulse window is chosen wide enough that $E(t)$ is negligible at the temporal boundaries of propagation.

TDSE propagation

The TDSE is propagated either (i) in an eigenstate expansion using the coupled-channel equations Eq. (15) with a unitary exponential propagator, or (ii) on a direct spatial grid with a unitary/symplectic scheme. Convergence is verified by reducing Δt and increasing the basis size (or grid resolution) until $d(t)$ and $P_{\nu m}(t)$ are stable.

Spectral and time–frequency analysis

Dipole spectra are obtained from the Fourier transform of the dipole moment, Eq. (28). Time–frequency maps are computed via a windowed transform, Eq. (29), with the gate width chosen to balance temporal and spectral resolution.

Statistics and reproducibility

All simulations reported in this manuscript are deterministic solutions of the TDSE for specified geometry and pulse parameters. Numerical reproducibility is established by convergence tests with respect to time step Δt, basis size N (or grid resolution), and absorber (CAP) settings, as described above [3].

Data availability

All data supporting the findings of this study are available within the article and its Supplementary Information. Source data for figures will be provided upon request.

Code availability

Custom code used for numerical propagation and post-processing is available from the corresponding author upon reasonable request.

Figure 10: Case IV: Quantum-number-resolved dynamics. Example of m- (and/or π_z-) resolved populations or dipole contributions under different polarizations, demonstrating symmetry-driven channel selectivity in the toroidal geometry.

Table 2: Example simulation parameter list (edit with your chosen values).

Category	Parameter	Meaning
Geometry	R, a	Torus major/minor radii
Confinement	V_0	Finite-barrier height (if used)
Pulse	E_0	Peak field amplitude
Pulse	ω_0	Carrier frequency
Pulse	τ	FWHM duration
Pulse	ϕ_{CEP}	Carrier-envelope phase
Numerics	Δt	Time step
Numerics	N	Number of eigenstates (basis method)
Numerics	η, layer width	CAP strength and width (optional)

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Author contributions

K.A. conceived the idea and performed the primary theoretical development. V.S.B. contributed to the modelling strategy, interpretation and manuscript preparation. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Extended data

Extended Data figures and tables (if any) are provided in the Supplementary Information.

Figure legends

Figure 1. Schematic of the torus geometry and coordinate definitions.
Figure 2. Case I: density evolution inside the torus during attosecond excitation.
Figure 3. Case I: dipole response and spectrum for the baseline parameters.
Figure 4. Case II: confinement-induced spectral shifts for different (R, a).
Figure 5. Case III: pulse-controlled channel populations $P_{\nu m}(t)$.
Figure 6. Case IV: quantum-number-resolved (m and π_z) dynamics.

Table legends

Table 1. Simulation parameters used for the torus-confined hydrogen calculations.

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